

EV Charging & Grid Impact

Hands-on with SimaaS · SUMO × Controller × PandaPower

In cooperation with

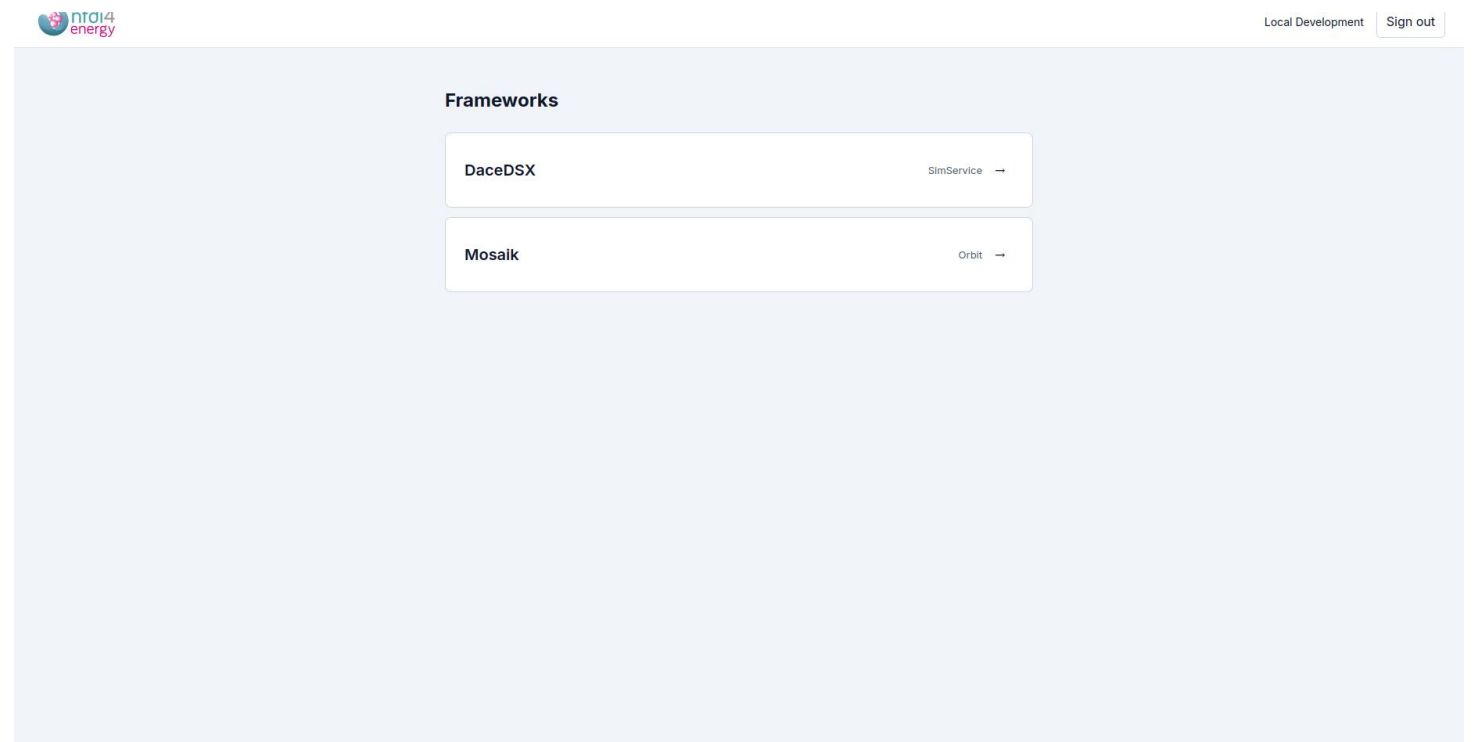


Part 1 — Getting Started

Open the Dashboard

Open the SimaaS start page.

Under Frameworks, select DaceDSX.



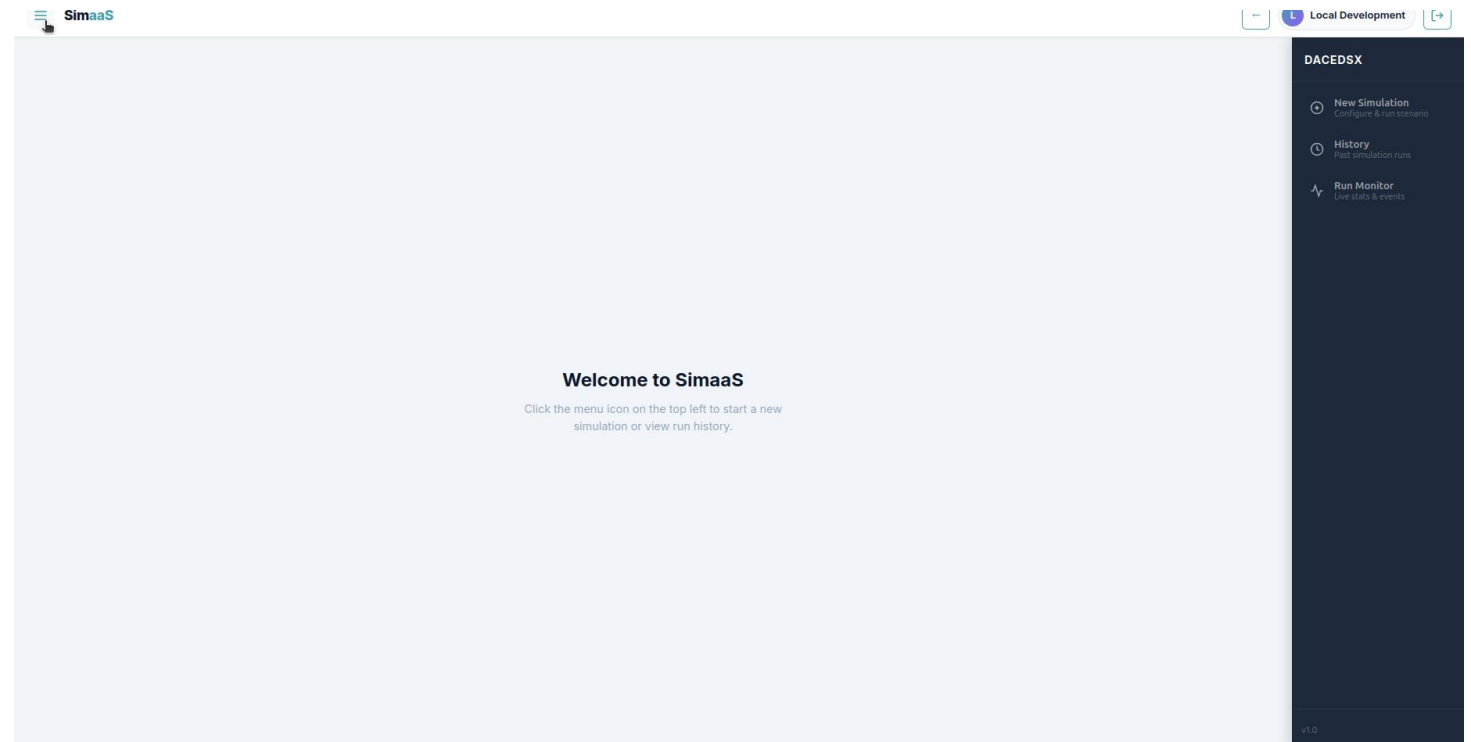
Find Your Way Around

Click the menu icon in the top left.

New Simulation → create a new scenario

History → past runs & results

Run Monitor → live view of a run



Part 2 — Build Your Scenario

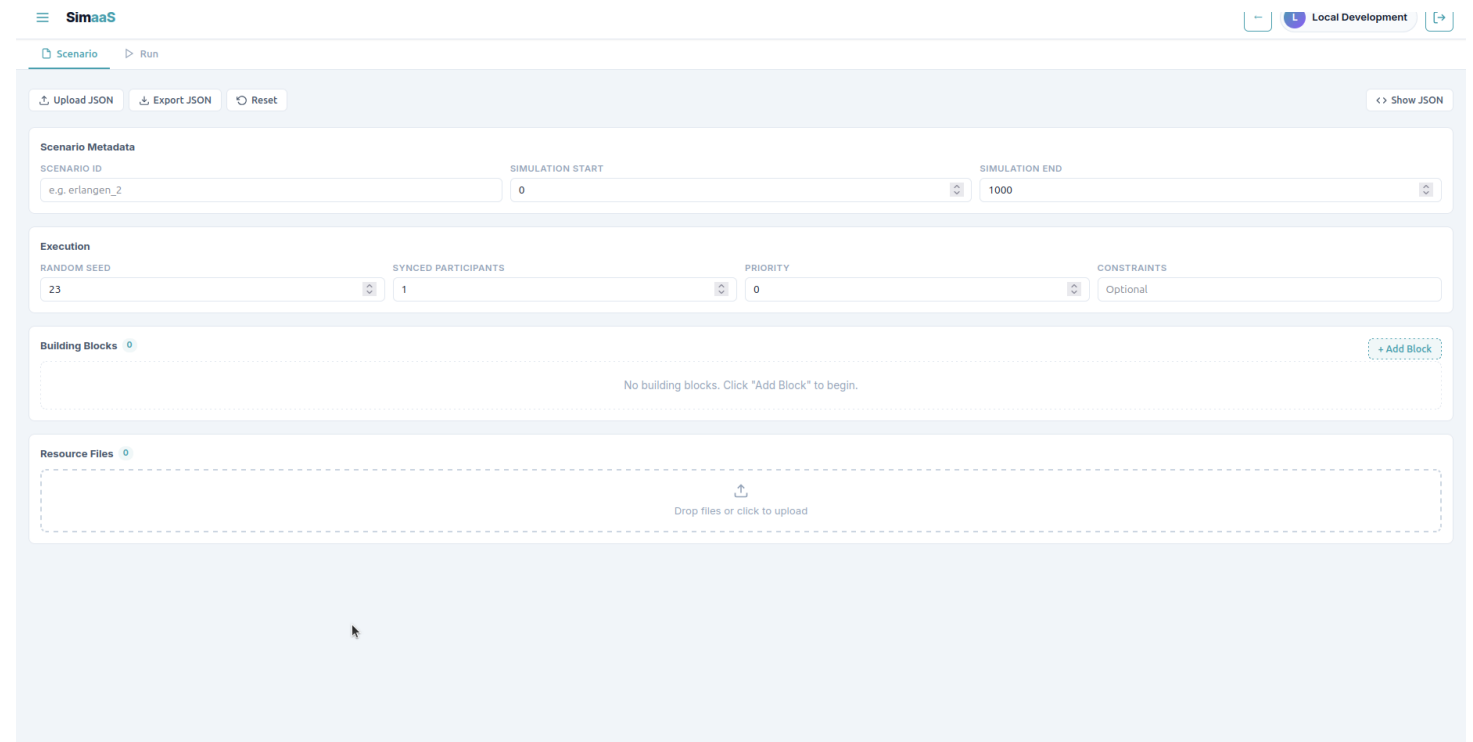
Create a New Scenario

Give it a Scenario ID, e.g. workshop_<your name>.

You can leave Simulation Start and Execution at their defaults.

For the use cases we want to run
Simulation End should be set to 3600.

Click + Add Block.



The screenshot shows the 'SimaaS' web interface for creating a new scenario. The top navigation bar includes a hamburger menu, the 'SimaaS' logo, and a 'Local Development' toggle. Below the navigation bar, there are tabs for 'Scenario' and 'Run'. The main form area contains several sections: 'Scenario Metadata' with fields for 'SCENARIO ID' (pre-filled with 'e.g. erlangen_2'), 'SIMULATION START' (0), and 'SIMULATION END' (1000); 'Execution' with fields for 'RANDOM SEED' (23), 'SYNCED PARTICIPANTS' (1), 'PRIORITY' (0), and 'CONSTRAINTS' (Optional); 'Building Blocks' with a dashed box and a '+ Add Block' button; and 'Resource Files' with a dashed box and a 'Drop files or click to upload' button. At the top right of the form, there are buttons for 'Upload JSON', 'Export JSON', 'Reset', and 'Show JSON'.

Building Block 1: SUMO

To run a SUMO Simulation you need to fill the following fields:

Instance ID: <can be anything you want>

Type: SumoWrapper

Step Length: 1000

Domain: traffic

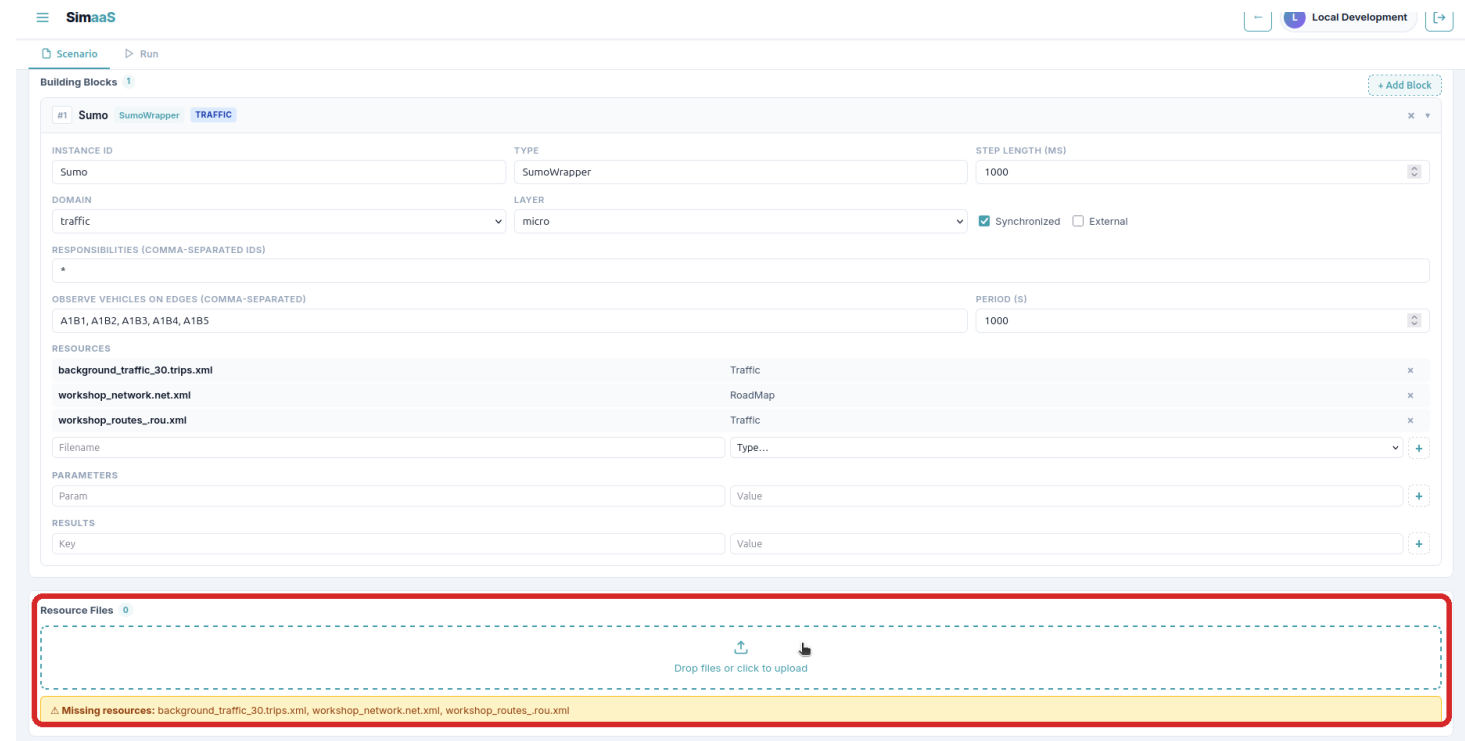
Layer: micro

Responsibilities: *

Observe vehicles on edges:

A1B1, A1B2, A1B3, A1B4, A1B5

We also need to add the resources with the corresponding type → see next slide.



SimaaS

Scenario Run

Building Blocks

#1	Sumo	SumoWrapper	TRAFFIC
INSTANCE ID	Sumo	SumoWrapper	STEP LENGTH (MS)
DOMAIN	traffic	LAYER	micro
RESPONSIBILITIES (COMMA-SEPARATED IDS)	*		
OBSERVE VEHICLES ON EDGES (COMMA-SEPARATED)	A1B1, A1B2, A1B3, A1B4, A1B5		PERIOD (S)
RESOURCES			
background_traffic_30.trips.xml		Traffic	x
workshop_network.net.xml		RoadMap	x
workshop_routes_rou.xml		Traffic	x
PARAMETERS			
Param		Value	+
RESULTS			
Key		Value	+

Resource Files

Drop files or click to upload

Missing resources: background_traffic_30.trips.xml, workshop_network.net.xml, workshop_routes_rou.xml

List and Upload the Resources

You need to add the following resources:

Filename: background_traffic_30.trips.xml

Type: Traffic

Filename: workshop_network.net.xml

Type: RoadMap

Filename: workshop_routes_.rou.xml

Type: Traffic

Upload the three provided files:

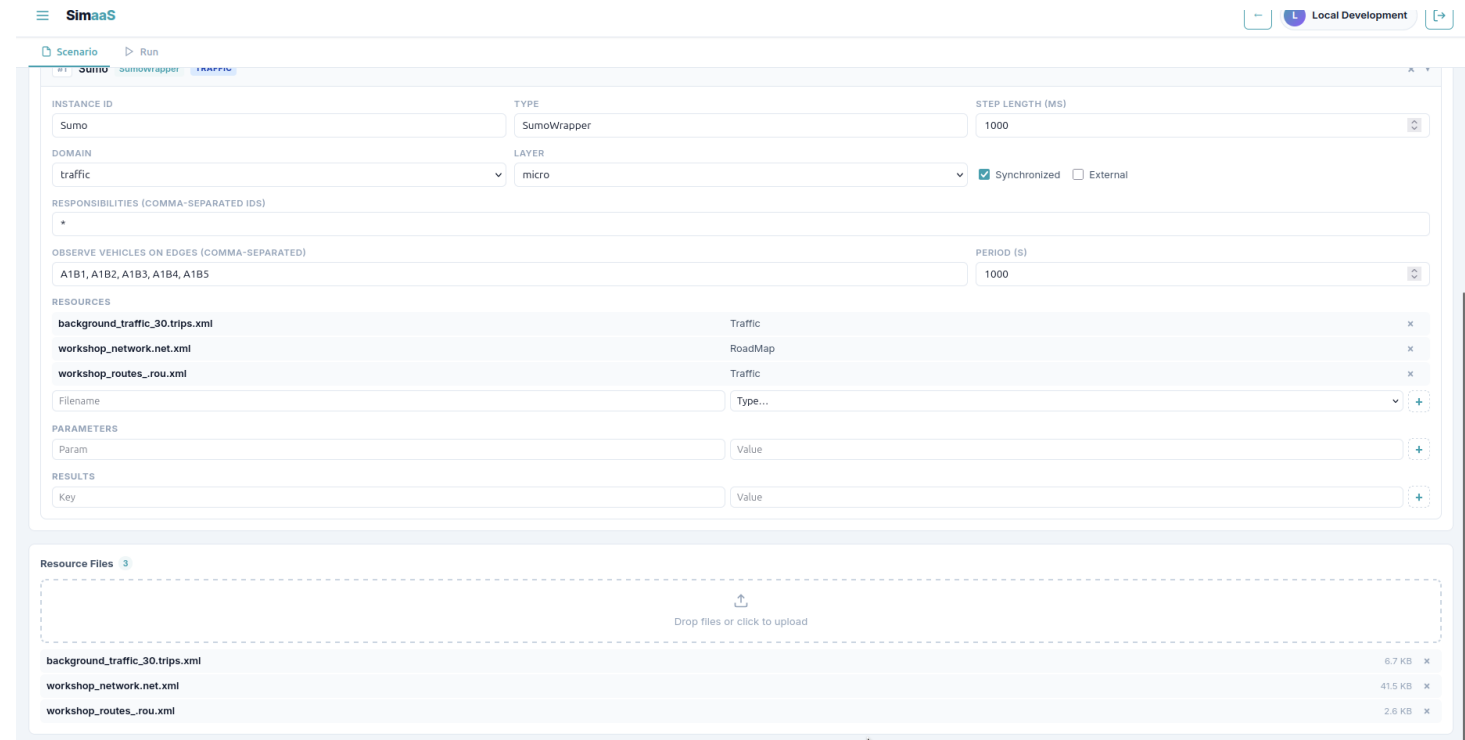
background_traffic_30.trips.xml

workshop_network.net.xml

workshop_routes_.rou.xml

The warning at the bottom disappears automatically as soon as you upload all the files that are listed.

If you uploaded everything you can run a simulation or continue adding other building blocks.



The screenshot shows the SimaaS web interface. At the top, there's a navigation bar with 'SimaaS' and a 'Local Development' button. Below the navigation bar, there's a 'Scenario' tab and a 'Run' button. The main area is divided into several sections:

- INSTANCE ID:** Sumo
- TYPE:** SumoWrapper
- STEP LENGTH (MS):** 1000
- DOMAIN:** traffic
- LAYER:** micro
- Synchronized:** ☒ **External:** ☐
- RESPONSIBILITIES (COMMA-SEPARATED IDS):** *
- OBSERVE VEHICLES ON EDGES (COMMA-SEPARATED):** A1B1, A1B2, A1B3, A1B4, A1B5
- PERIOD (S):** 1000
- RESOURCES:**
 - background_traffic_30.trips.xml (Traffic)
 - workshop_network.net.xml (RoadMap)
 - workshop_routes_.rou.xml (Traffic)
- PARAMETERS:** A table with 'Param' and 'Value' columns.
- RESULTS:** A table with 'Key' and 'Value' columns.
- Resource Files:** A section showing the uploaded files: background_traffic_30.trips.xml (6.7 KB), workshop_network.net.xml (41.5 KB), and workshop_routes_.rou.xml (2.6 KB).

Building Block 2: PandaPower

Instance ID: <can be anything you want>

Type: PandaPowerWrapper

Step Length: 50000

Domain: energy

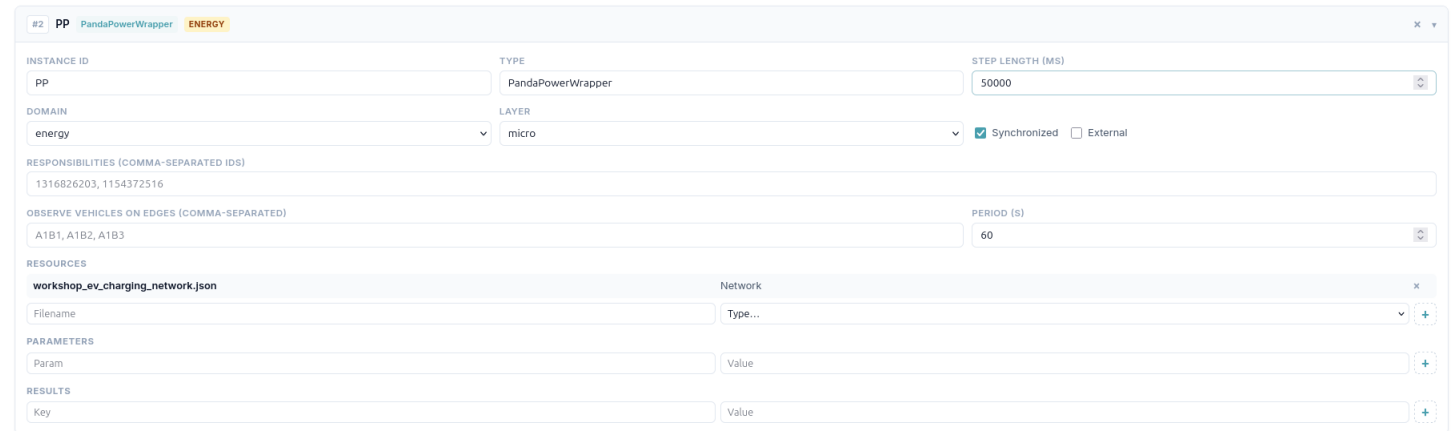
Layer: micro

Resource:

Fieldname:
workshop_ev_charging_network.json

Type: Network

Don't forget to upload the resource as well.



The screenshot shows the configuration interface for the PandaPowerWrapper. The interface is divided into several sections:

- INSTANCE ID:** A text input field containing "PP".
- TYPE:** A dropdown menu set to "PandaPowerWrapper".
- STEP LENGTH (MS):** A numeric input field set to "50000".
- DOMAIN:** A dropdown menu set to "energy".
- LAYER:** A dropdown menu set to "micro".
- Synchronization:** Two checkboxes, "Synchronized" (checked) and "External" (unchecked).
- RESPONSIBILITIES (COMMA-SEPARATED IDS):** A text input field containing "1316826203, 1154372516".
- OBSERVE VEHICLES ON EDGES (COMMA-SEPARATED):** A text input field containing "A1B1, A1B2, A1B3".
- PERIOD (S):** A numeric input field set to "60".
- RESOURCES:** A section with a table-like structure. It has a header row with "workshop_ev_charging_network.json" and "Network". Below it, there are columns for "Filename" and "Type...", with a "+" button to add more resources.
- PARAMETERS:** A section with a table-like structure. It has a header row with "Param" and "Value", with a "+" button to add more parameters.
- RESULTS:** A section with a table-like structure. It has a header row with "Key" and "Value", with a "+" button to add more results.

Building Block 3: Controller

Instance ID: <can be anything you want>

Type: ControllerWrapper

Step Length: 50000

Resources:

Fieldname: station_mapping.json

Type: Config

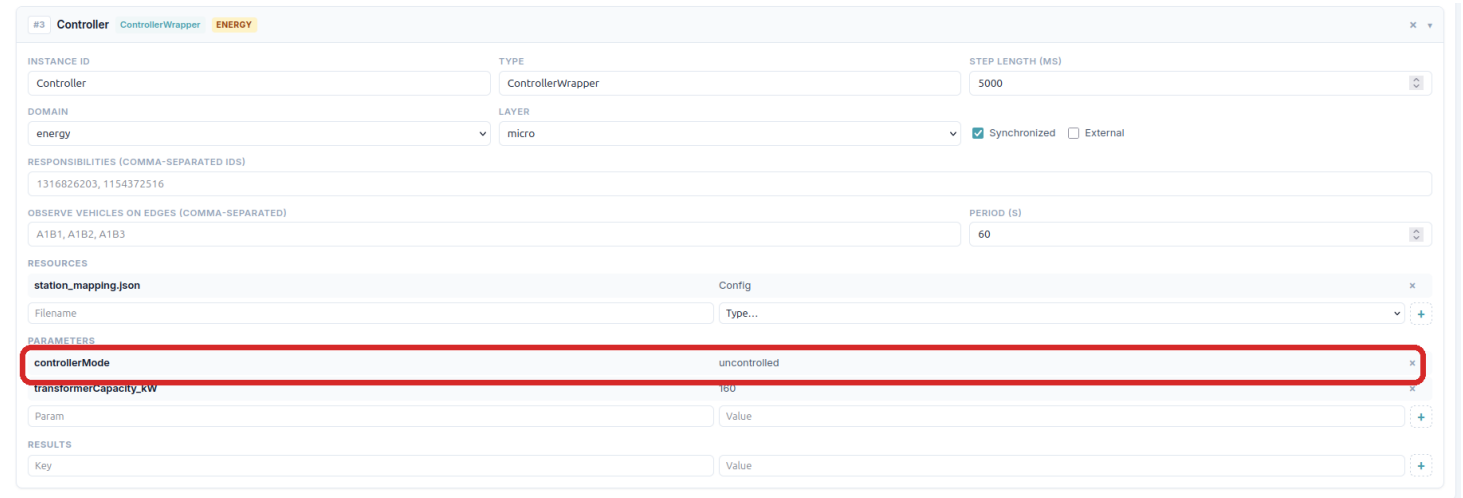
Important: We need to add two parameters!

transformerCapacity_kW = 160

controllerMode = uncontrolled

You'll change exactly this field again shortly.

Don't forget to upload the resource as well.



The screenshot shows the configuration interface for a Controller instance. The instance is named "Controller" and is of type "ControllerWrapper". The step length is set to 5000. The domain is "energy" and the layer is "micro". The responsibilities are "1316826203, 1154372516". The observed vehicles on edges are "A1B1, A1B2, A1B3" with a period of 60 seconds. The resources section shows "station_mapping.json" as a Config resource. The parameters section is highlighted with a red box and contains two parameters: "controllerMode" with value "uncontrolled" and "transformerCapacity_kW" with value "160".

INSTANCE ID	TYPE	STEP LENGTH (MS)
Controller	ControllerWrapper	5000

DOMAIN	LAYER	☒ Synchronized ☐ External
energy	micro	

RESPONSIBILITIES (COMMA-SEPARATED IDS)

1316826203, 1154372516

OBSERVE VEHICLES ON EDGES (COMMA-SEPARATED)

A1B1, A1B2, A1B3

PERIOD (s)

60

RESOURCES	Type
station_mapping.json	Config

PARAMETERS	Value
controllerMode	uncontrolled
transformerCapacity_kW	160

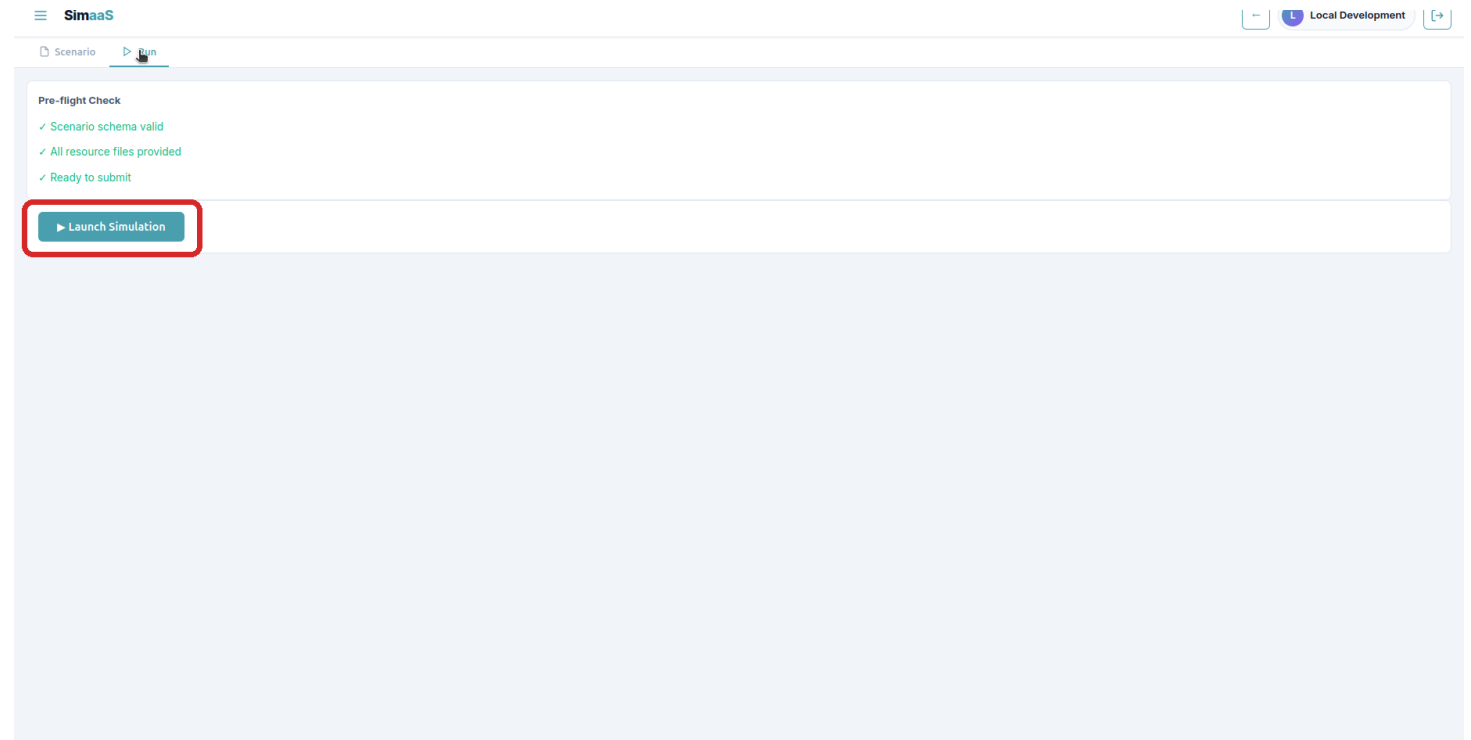
RESULTS	Value
Key	Value

Part 3 — Run & Observe

Pre-flight Check & Launch

Switch to the Run tab.

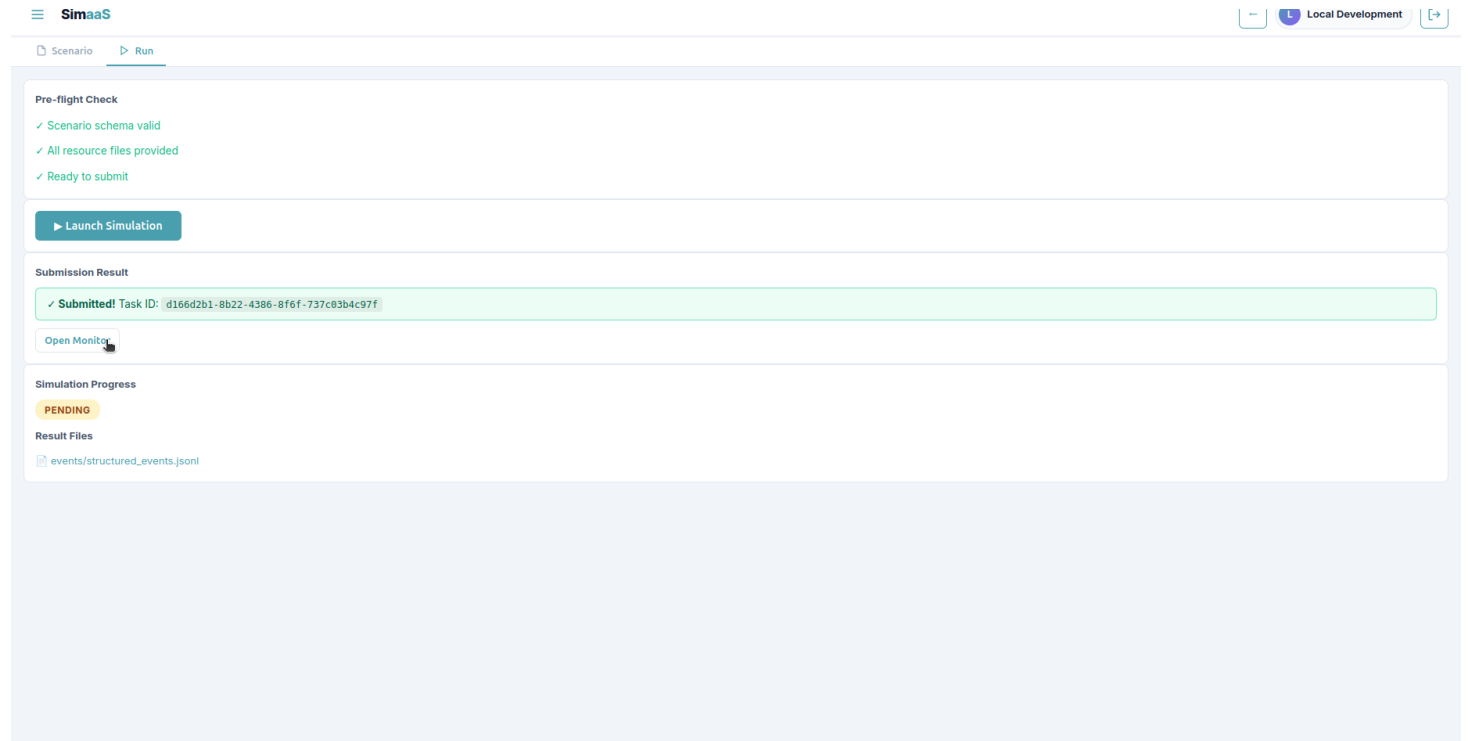
Once all three checks are green,
click Launch Simulation.



Confirm Your Submission

You'll receive a Task ID.

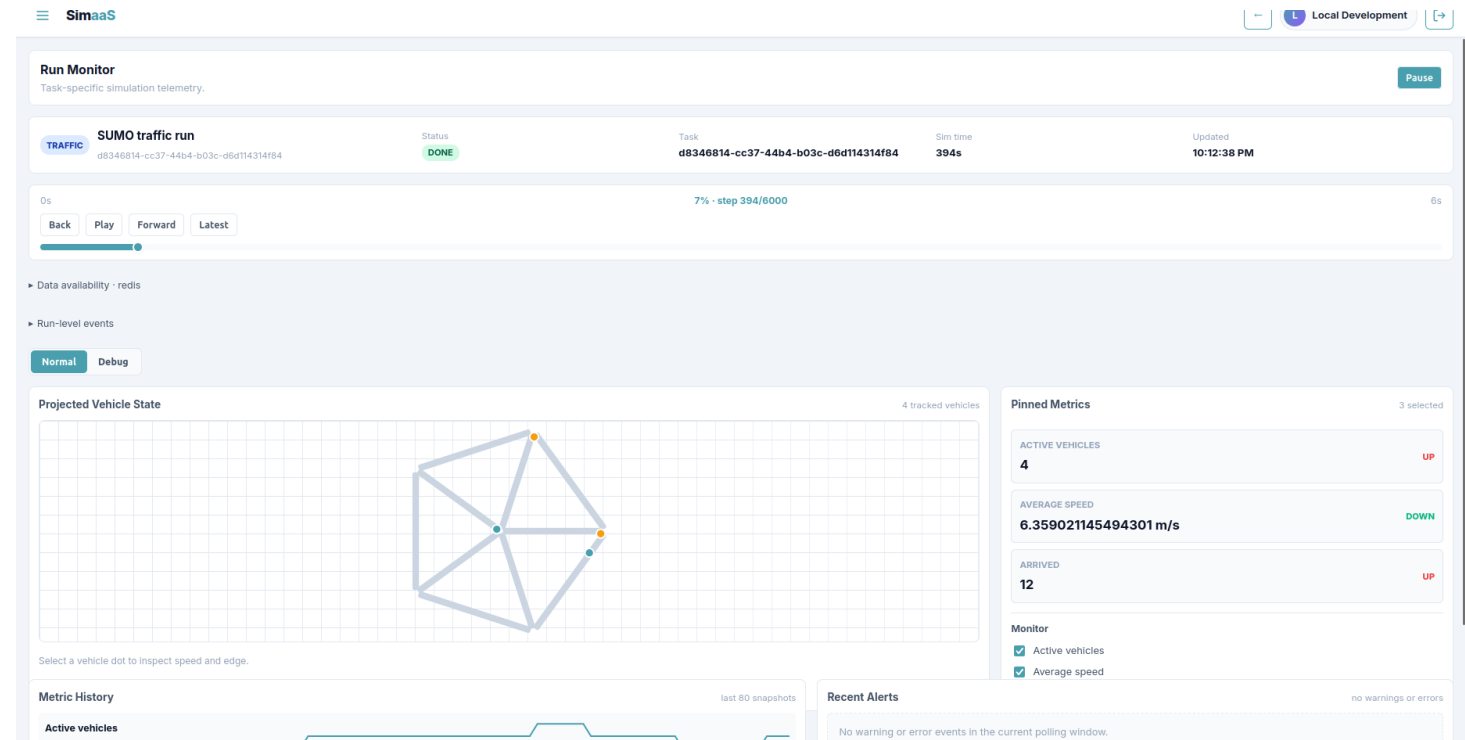
Click Open Monitor
to watch it live.

A screenshot of the SimaaS web interface. The top navigation bar includes a hamburger menu, the "SimaaS" logo, and a "Local Development" toggle. Below the navigation bar, there are two tabs: "Scenario" and "Run", with "Run" being the active tab. The main content area is divided into three sections. The first section, "Pre-flight Check", contains three green checkmarks: "✓ Scenario schema valid", "✓ All resource files provided", and "✓ Ready to submit". Below this is a blue button labeled "▶ Launch Simulation". The second section, "Submission Result", features a green bar with the text "✓ Submitted! Task ID: d166d2b1-8b22-4386-8f6f-737c03b4c97f". Below this bar is a button labeled "Open Monitor". The third section, "Simulation Progress", shows a yellow "PENDING" status. Below this, under the heading "Result Files", there is a link to "events/structured_events.json".

Watch It Live

Run Monitor shows progress, active vehicles, and the simulated map in real time.

If you change the order of the building blocks and list the PandaPowerWrapper first you will see a view of the electric network.



Part 4 — Compare Controllers

Before You Continue: Your Prediction

- **What happens when every vehicle charges at full power, uncontrolled?**
 - ☐ No noticeable impact
 - ☐ Increased transformer loading
 - ☐ Transformer overload
 - ☐ Other: _____
-

Only ONE Value Changes

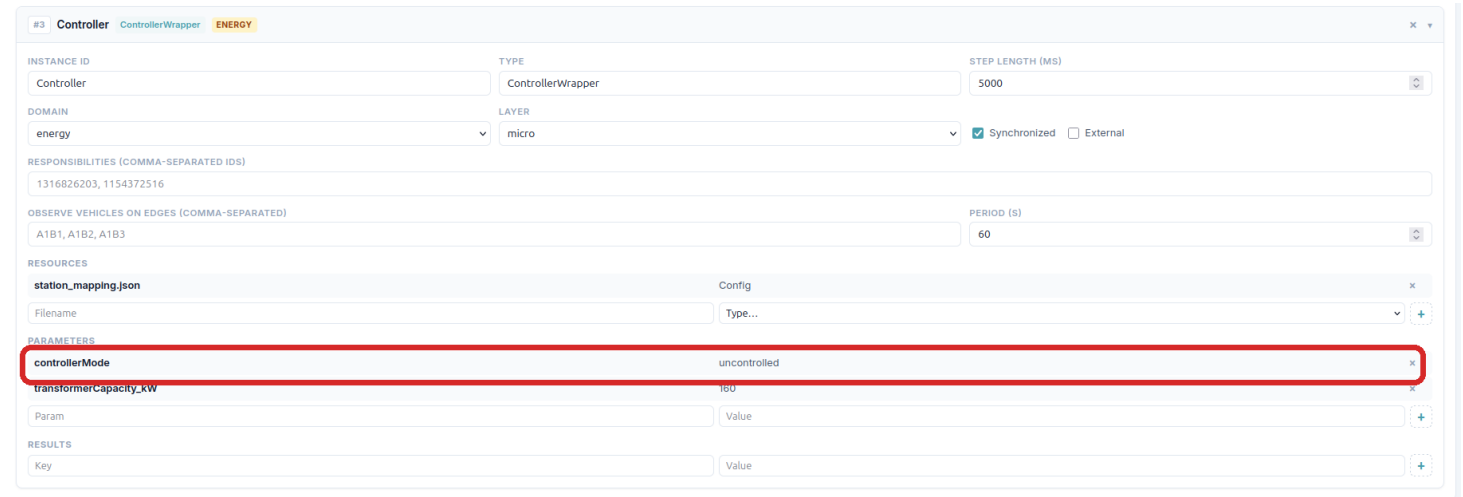
Go back to the Controller block.

Change controllerMode:

uncontrolled → safe_mode

Everything else stays exactly the same.

Run it again (Part 3).

A screenshot of the nfdi4energy software interface showing the configuration for a 'Controller' block. The interface is divided into several sections: INSTANCE ID, TYPE, STEP LENGTH (MS), DOMAIN, LAYER, RESPONSIBILITIES (COMMA-SEPARATED IDS), OBSERVE VEHICLES ON EDGES (COMMA-SEPARATED), PERIOD (S), RESOURCES, PARAMETERS, and RESULTS. The 'controllerMode' parameter is highlighted with a red rectangle, showing its current value as 'uncontrolled'. Other parameters like 'transformerCapacity_kW' are also visible.

INSTANCE ID	TYPE	STEP LENGTH (MS)
Controller	ControllerWrapper	5000

DOMAIN	LAYER	☒ Synchronized ☐ External
energy	micro	

RESPONSIBILITIES (COMMA-SEPARATED IDS)
1316826203, 1154372516

OBSERVE VEHICLES ON EDGES (COMMA-SEPARATED)
A1B1, A1B2, A1B3

PERIOD (S)
60

PARAMETERS	
station_mapping.json	
Filename	Type...
controllerMode	uncontrolled
transformerCapacity_kW	160
Param	Value

RESULTS	
Key	Value

Find Your Results

Open History.

Download for your run:

trafo_p_mw.csv

charging_log.csv

SimaaS

Local Development

Simulation History

SCENARIO

TASK ID

SUBMITTED

STATUS

scenario16	d166d2b1...	20 Sept 2026 08:29	PENDING
scenario_heuristic	c8900a92...	19 Sept 2026 20:21	DONE
FULL TASK ID c8900a92-0813-4397-9cd7-678e5b1a0778 RESULT FILES bus_va_degree.csv bus_vm_pu.csv charging_log.csv events/metrics.jsonl events/structured_events.jsonl history.json line_p_from_mw.csv logs/measurements_scenario_heuristic_Sumo.csv logs/measurements_scenario_heuristic_Sumo_summary.csv logs/measurements_scenario_heuristic_Sumo_vehs.txt trafo_p_mw.csv Open Monitor			
scenario_safe_mode	8be6d59f...	19 Sept 2026 20:21	DONE
scenario25	d8346814...	19 Sept 2026 20:12	DONE
scenario16	cd7e93d0...	19 Sept 2026 20:11	PENDING
scenario23	b5c6fcfd...	19 Sept 2026 20:10	DONE
scenario22	49a5a409...	19 Sept 2026 19:55	ERROR
scenario16	4f1f99c8...	19 Sept 2026 19:54	ERROR
scenario20	07dc6cf...	19 Sept 2026 19:53	RUNNING
scenario16	f1da1651...	19 Sept 2026 19:48	PENDING

DACEDSX

New Simulation

History

Run Monitor

v1.0

Plot Your Results

- We prepared a little Python script for you if you want to plot your results do the following steps:
 - `pip install matplotlib`
 - Save the result files `trafo_p_mw.csv` and `charging_log.csv` to a folder (e.g. `uncontrolled`)
 - Run the python script:
 - `python plot_results.py --run uncontrolled`
 - You can save the results of the other runs in separate folders and plot everything at the same time:
 - `python plot_results.py --run uncontrolled/ --run safe_mode/ --run heuristic/`
-

Comparing the Three Strategies

	Uncontrolled	Safe Mode	Heuristic
Transformer Peak (kW)	?	?	?
Fully Charged	?	?	?
Grid Violation	?	?	?

What's the Trade-off?

- Not: which mode is “better”?
 - But: what effect does
 - each strategy have?
-

Wrap-up & Feedback

Your Feedback

- 👍 **What worked well?**
 - 🤔 **What was unclear or difficult?**
 - 💡 **What would you improve?**
 - 🔧 **What would you need to use a workflow like this for your own research?**
-

Contact

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